

Statistical analysis of experimental data

Introduction to Python

Artur Kalinowski (for Aleksander Filip Żarnecki)



Lecture 01

October 8, 2026

Introduction to Python

- 1 Introduction
- 2 Environment
- 3 Python packages
- 4 Homework

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Statistical analysis of experimental data

The goal of this lecture is to present basic concepts and methods of statistical analysis of experimental data...

We will quite often refer to problems encountered in the elementary particle physics, as this is the perfect test ground for statistical analysis

This is because all particles (of the same type) are identical.

We can easily assure same initial conditions for the repeated measurement...

This is often a fundamental problem in social or medical sciences...

The knowledge of particle physics is not required (but for some fundamental concepts like the invariant mass or mean lifetime) and the presented analysis methods can clearly be used also in other fields of science...

Statistical analysis of experimental data

The goal of the lecture is not only to present the theoretical concepts but also to demonstrate how they can be applied in practical terms to data analysis.

Different concepts as well as analysis methods will be discussed using code examples.

We will address many (relatively simple) problems and try to look at their numerical solutions.

All examples presented are based on Python and multiple additional packages, which will be shortly introduced today.

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The focus of the course is on understanding fundamental principles.

Examples will often be simplified and code examples not very efficient.

The main goal is to develop your intuition regarding the statistical interpretation of data!

Course plan

14 lectures, Thursdays, 9:15 – 12:00, room B2.38 (with two breaks)
From October 8th, 2026 to January 21th, 2027

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Homework exercises

Solutions have to be uploaded to Kampus within two weeks!

Until midnight before the lecture! Proposed solutions will be presented at the lecture.

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Assessment criteria

Minimum of 50% of points collected from exercises and exam (with the same weight).

Assessment in September: 50% of points from written examination only.

Some references

- G. Bohm and G. Zech, Introduction to Statistics and Data Analysis for Physicists, Verlag Deutsches Elektronen-Synchrotron, [3rd edition](#);
- S. Brandt, Data Analysis: Statistical and Computational Methods for Scientists and Engineers, Springer 2014;
- **M. Bonamente, Statistics and Analysis of Scientific Data**, Springer 2017;
- R.J. Barlow, Practical Statistics for Particle Physics, [PDF from arXiv](#);
- Max Bramer, Principles of Data Mining, Springer 2016;
- David Forsyth, Probability and Statistics for Computer Science, Springer 2018;
- [Particle Physics Reference Library Volume 2](#) (chapter 15), Springer 2020.

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Web resources

Kampus platform will be used for lecture slides, Python notebooks, as well as home exercises (and final exam):

<https://kampus-kursy.ckc.uw.edu.pl/course/view.php?id=10840>

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All scripts discussed at the lecture will be available for download from the github repository:

<https://github.com/zarnecki/SAED>

Inserted in the lecture slides are links to open them in your browser using the Google Colab.

You can also use this link  [Open in Colab](#) to open repository and select the notebook to load.

Please remember that you need to save the file to (your) Google Disc before you make any modifications. Otherwise, all your changes will be lost!..

JupyterLab

All scripts discussed during the course were developed with [Jupyter Lab](#)
Jupyter's tools are available for installation via the Python Package Index (pip)

Containers

If you want to avoid installing and configuring all the required packages on your personal computer, you can also use virtual environments known as containers (**lightweight versions of old time virtual machines**). Two popular environments used to run the containers:

- [Apptainer](#)
- [Docker](#)

Both Docker and Apptainer can use the same input files.

Docker is available for any operating system, Apptainer only for Linux ones.

Notebooks presented at this lecture can be run with [akalinow/root-fedora39](#) container prepared for the other course some time ago. This is a container based on Fedora Linux ditribution, and contains all packages we will use during the classes.

Python tools

One of the main advantages of Python is the large number of diverse packages developed for various applications...

During the course we will use the following packages for computation and plotting:

- [numpy](#) - The fundamental package for scientific computing with Python
- [matplotlib](#) - Visualization with Python
- [SciPy](#) - Fundamental algorithms for scientific computing in Python
- [scikit-learn](#) - Machine Learning in Python, simple and efficient
- [pandas](#) - Handling more complex data structures

See lecture notebook for more details and examples:



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NumPy documentation

Version: 2.0.dev0

Download documentation: [Historical versions of documentation](#)

Useful links: [Installation](#) | [Source Repository](#) | [Issue Tracker](#) | [Q&A Support](#) | [Mailing List](#)

NumPy is the fundamental package for scientific computing in Python. It is a Python library that provides a multidimensional array object, various derived objects (such as masked arrays and matrices), and an assortment of routines for fast operations on arrays, including mathematical, logical, shape manipulation, sorting, selecting, I/O, discrete Fourier transforms, basic linear algebra, basic statistical operations, random simulation and much more.



Getting Started

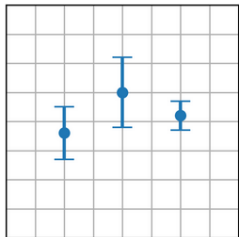


User Guide

Matplotlib: Visualization with Python

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

- Create [publication quality plots](#).
- Make [interactive figures](#) that can zoom, pan, update.
- Customize [visual style and layout](#).
- Export to [many file formats](#).
- Embed in [JupyterLab](#) and [Graphical User Interfaces](#).
- Use a rich array of [third-party packages](#) built on Matplotlib.

[Try Matplotlib \(on Binder\)](#)

`errorbar(x, y, yerr, xerr)`

[Getting Started](#)[Examples](#)[Reference](#)[Cheat Sheets](#)[Documentation](#)

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Fundamental algorithms for scientific computing in Python

GET STARTED

SciPy 1.11.3 released! 2023-09-27

FUNDAMENTAL ALGORITHMS

SciPy provides algorithms for optimization, integration, interpolation, eigenvalue problems, algebraic equations, differential equations, statistics and many other classes of problems.

BROADLY APPLICABLE

The algorithms and data structures provided by SciPy are broadly applicable across domains.

pandas documentation

Date: Sep 20, 2023 **Version:** 2.1.1

Download documentation: [Zipped HTML](#)

Previous versions: Documentation of previous pandas versions is available at pandas.pydata.org.

Useful links: [Binary Installers](#) | [Source Repository](#) | [Issues & Ideas](#) | [Q&A Support](#) | [Mailing List](#)

pandas is an open source, BSD-licensed library providing high-performance, easy-to-use data structures and data analysis tools for the [Python](#) programming language.



Getting started

New to *pandas*? Check out the getting started guides.



User guide

The user guide provides in-depth information on the



Install User Guide API Examples Community More ▾

 Go

scikit-learn

Machine Learning in Python

Getting Started

Release Highlights for 1.3

GitHub

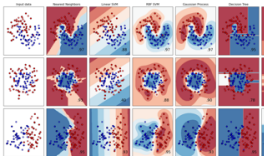
- Simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.

Algorithms: Gradient boosting, nearest neighbors, random forest, logistic regression, and more...



Examples

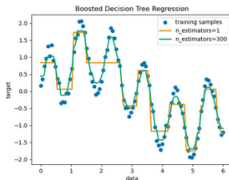
A.Kalinowski

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, Stock prices.

Algorithms: Gradient boosting, nearest neighbors, random forest, ridge, and more...



Examples

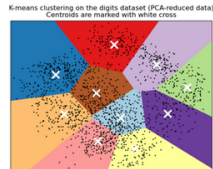
Statistical analysis 01

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, Grouping experiment outcomes

Algorithms: k-Means, HDBSCAN, hierarchical clustering, and more...



Examples

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Graphical analysis of the data

Data set for this homework was selected from historical demographic data available from [OECD Data Explorer](#). The [CSV file](#) accessible from the web includes data describing age and gender structure of the population of Poland, as estimated in years from 1950 till 2025.

- 1 Prepare the plot showing the population of men and women born in 1946-1950 (not older than 4 in 1950), as a function of year (or age).
- 2 Try to estimate median life expectancy (the age at which exactly 50% of the initial population remains) for men and women born in 1946-1950. What problems do you encounter in your estimate?

Example code included in the lecture notebook shows how to access the data set using pandas.

Solutions should be uploaded until October 21.

Problem solutions

Apply for all homework and exam problems.

Solutions should be uploaded to Kampus as PDF documents (scan/photo of hand-written solution is also OK) including description/justification of the approach used, numerical results and relevant plots, as well as result discussion.

The Python notebook can be uploaded as a second file or the link to notebook in Google Colab can be added as a comment.

Make sure your code runs “from scratch” (no hidden dependencies) and Google Colab accessible.

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By uploading the solutions to Kampus (homework and exam) you declare that they resulted from your own work and that you have not obtained, shared nor discussed them with anyone. This includes AI...